

Results Summary: Does Learned Pruning Beat Simple Heuristics?

GAT + RL edge-pruning for botnet containment in Social IoT graphs, benchmarked against a full structural-baseline suite. All numbers are means over 40 independent scale-free graph instances (Barabasi-Albert), with a feature-driven infection model so pruning risky edges measurably changes outcomes.

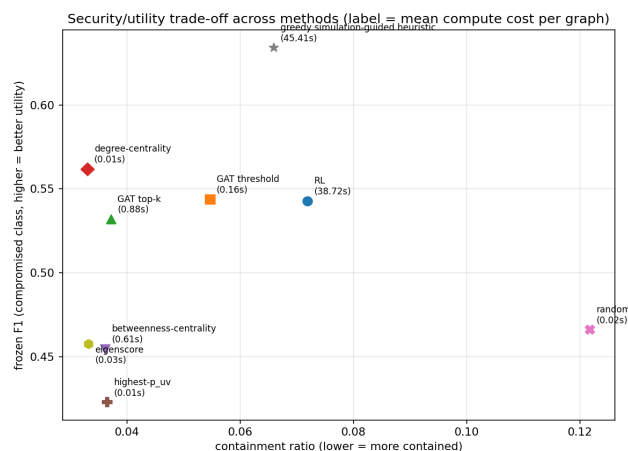
Headline finding

Simple degree-centrality pruning ties or beats the learned GAT+RL pipeline on containment, at a fraction of the cost. RL loses significantly (Holm-corrected $p < 0.05$) to every structural baseline tested -- including the spectral/eigenvalue method (eigenscore) the epidemic-containment literature treats as its principled standard. There is no single winner: it is a genuine containment-vs-utility trade-off (Pareto frontier), not a ranking with one method on top.

All nine methods compared (n = 40 graphs each)

Method	Containment ratio (mean +/- std)	Utility (frozen F1)	Compute cost /graph
Degree-centrality	0.0331 +/- 0.0061	0.562	0.0108 s
Eigenscore (spectral)	0.0332 +/- 0.0064	0.458	0.034 s
Betweenness-centrality	0.0361 +/- 0.0058	0.455	0.611 s
Highest-p_uv (direct risk)	0.0365 +/- 0.0050	0.423	0.008 s
GAT top-k	0.0372 +/- 0.0060	0.532	0.881 s
GAT threshold	0.0547 +/- 0.0126	0.544	0.161 s
Greedy simulation-guided	0.0659 +/- 0.0344	0.634	45.41 s
RL (adaptive policy)	0.0719 +/- 0.0430	0.543	38.72 s
Random	0.1218 +/- 0.0183	0.466	0.024 s

Sorted by containment ratio (lower = better, fewer nodes end up compromised). Green row = best containment. 14 pre-registered paired comparisons, Holm-Bonferroni corrected.



Key takeaways

- **Degree-centrality** wins on containment AND cost (about 3,600x cheaper than RL).
- **Greedy simulation-guided heuristic** wins on utility (frozen F1 0.634) but costs about 4,200x more than degree.
- **Eigenscore** (the field's spectral "gold standard") only ties degree -- doesn't beat it.
- **RL** beats only random pruning; ties the greedy heuristic; loses to every structural method.
- A myopia diagnostic (n=10 graphs) confirms the greedy heuristic's shortfall is structural (one-step-at-a-time), not a noisy search signal -- giving it 5x more search accuracy made results worse, not better.
- GAT+RL's containment result is architecture-conditional (only holds at heads=4, depth=2) but robust to feature noise/missingness.

Full numbers, statistical tests, ablations, and every caveat: [NOTES.md](#) and [harness_summary.csv](#) / [harness_paired_tests.csv](#) in the project repository.